

Reinforcement Learning 2024/2025

Asymmetric Self-Play for Automatic Goal Discovery in Robotic Manipulation

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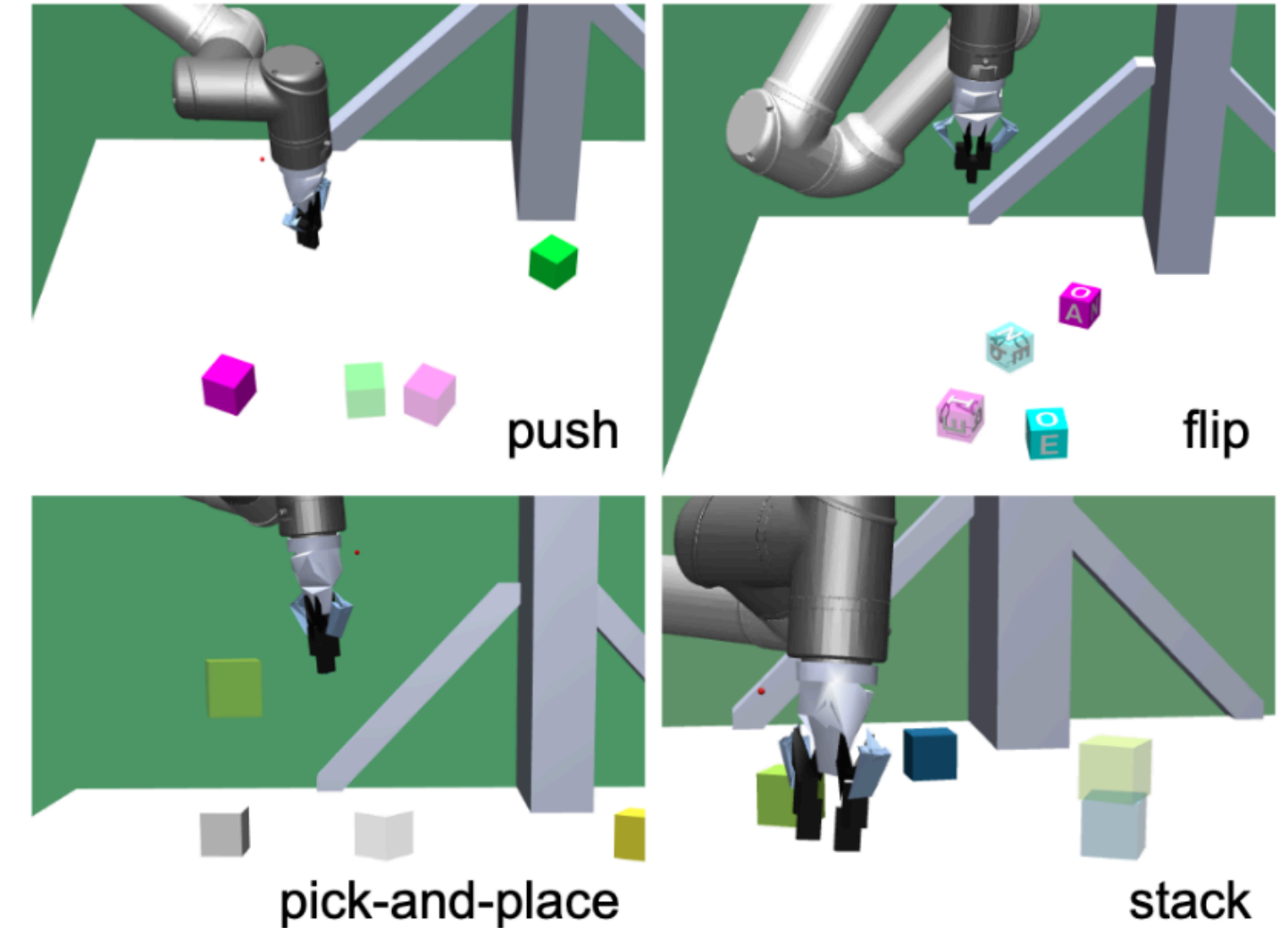
Based on the work of OpenAI et al. (2021), [arXiv:2101.04882](https://arxiv.org/abs/2101.04882)

The curriculum problem

How can we train a robot to solve a wide variety of tasks without manually defining every single one?

Challenge Creating a training distribution of goals that is **rich**, **diverse**, and **progressively difficult**, enabling *zero-shot generalization*.

Goal Learn to solve these tasks **without being explicitly trained on them**.



Traditional methods require extensive human effort for task design or demonstration, which doesn't scale for complex, open-ended manipulation.

Asymmetric Self-Play

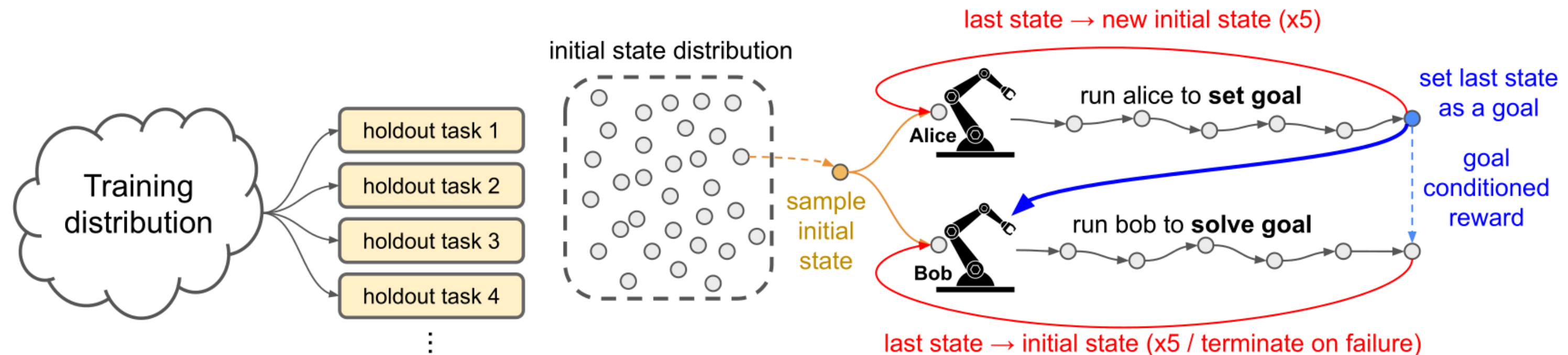
We automate goal generation through a **competitive game** between two agents.

Alice (goal setter)

- Plays with objects on the table.
- The final state becomes the **goal**.
- **Rewarded** if **Bob fails**, encouraging her to find challenging goals.

Bob (goal solver)

- A **goal-conditioned** policy.
- Must match the goal set by Alice.
- **Rewarded** for **success**.



Asymmetric Self-Play

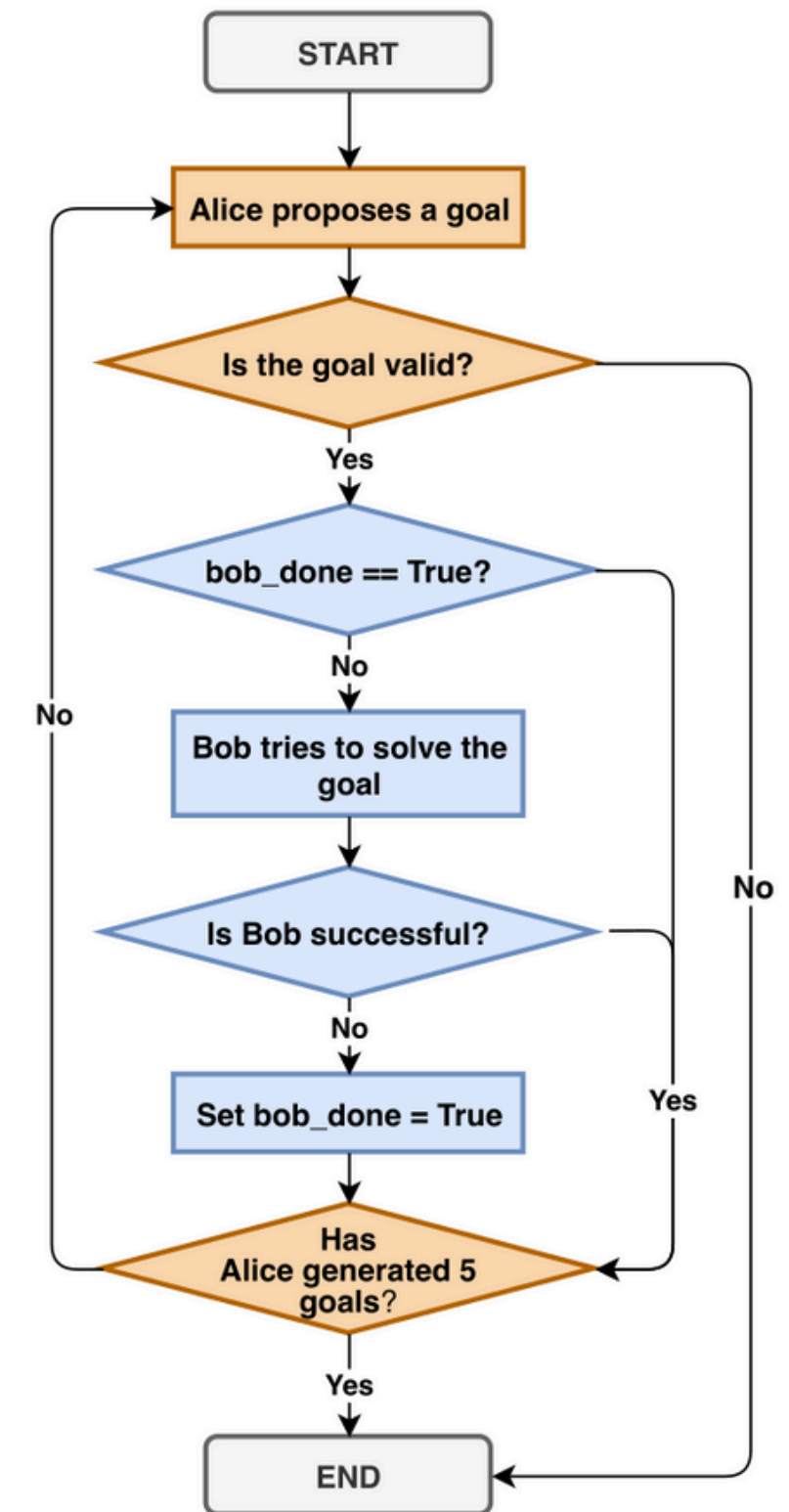
Alice and Bob engage in a **turn-based multi-goal game** within each episode.

Alice's turn Manipulates objects for a fixed number of steps; her final state becomes the goal.

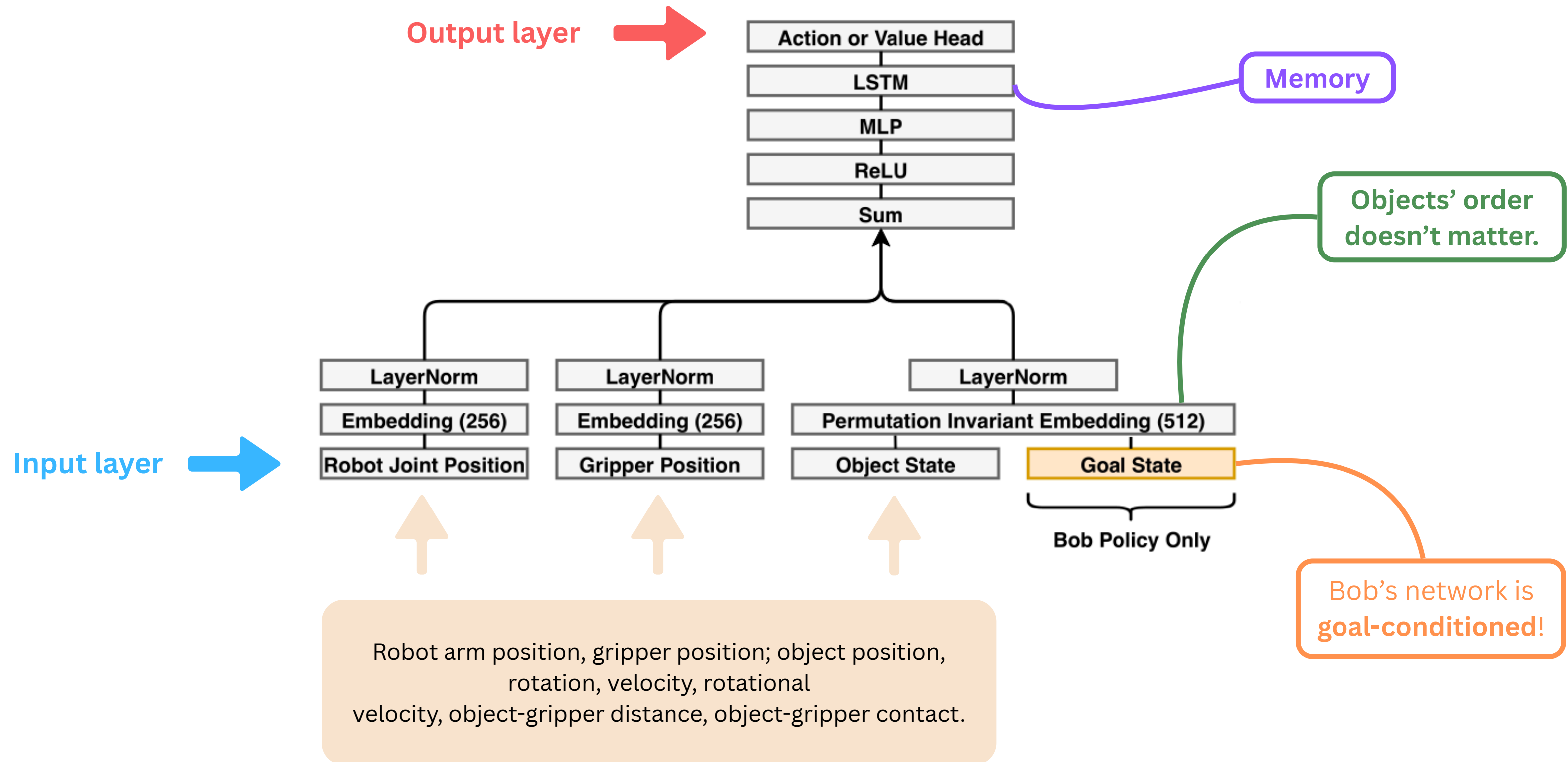
Bob's turn Attempts to achieve Alice's proposed goal, starting from the same initial state.

Competitive cycle

- Steps repeat up to 5 times per episode.
- If Bob fails, Alice continues setting goals.



What are Alice and Bob?



Setup

Both Alice and Bob are trained with *PPO* using *Adam* as optimizer.

Learning parameter	Value
discount factor	0.998
Generalized Advantage Estimation	0.95
entropy regularization coefficient	0.01
PPO clipping	0.2
ABC clipping	0.2
learning rate	3e-4
sample reuse	3
value loss weight	1.0
ABC loss weight	0.5

Parameter	Value
GPUs	1
rollout workers CPUs	12 x 1
batch size	4096
Alice's goal setting steps	100
Bob's maximum goal solving steps	200

Alice's and Bob's sparse rewards

The asymmetric reward structure drives a competitive "arms race" between the agents.

Alice	Bob	Alice reward	Bob reward
Invalid goal	-	0	-
Out-of-zone goal	Failure	$1 - 3 + 5$	$0 + \text{per-object-reward}$
Out-of-zone goal	Success	$1 - 3 + 0$	$5 + \text{per-object-reward}$
Valid goal	Failure	$1 + 5$	$0 + \text{per-object-reward}$
Valid goal	Success	$1 + 0$	$5 + \text{per-object-reward}$

Alice

- +5 **reward** if Bob **fails** to achieve her valid goal.
- +1 **reward** for setting any valid goal.
- -3 **penalty** if goal objects are placed out of the robot's workspace.

Bob

- +5 **reward** for successfully achieving the proposed goal.
- +1 **reward** per object placed correctly towards the goal.
- -1 **penalty** if an object is moved further from its goal state.

Alice Behavioral Cloning (ABC)

How does Bob learn complex tasks with only a sparse "success/fail" reward?

Problem For a difficult goal like stacking, random exploration will almost **never succeed**, so Bob gets no learning signal.

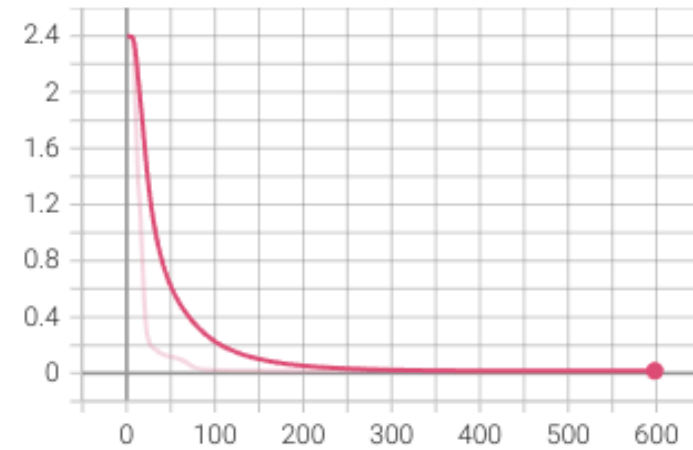
Insight Alice's own trajectory is a **successful demonstration** of how to reach the goal she created.

When Bob fails... we relabel Alice's trajectory as a **goal-conditioned demonstration**. Bob then learns from this via *behavioral cloning*.

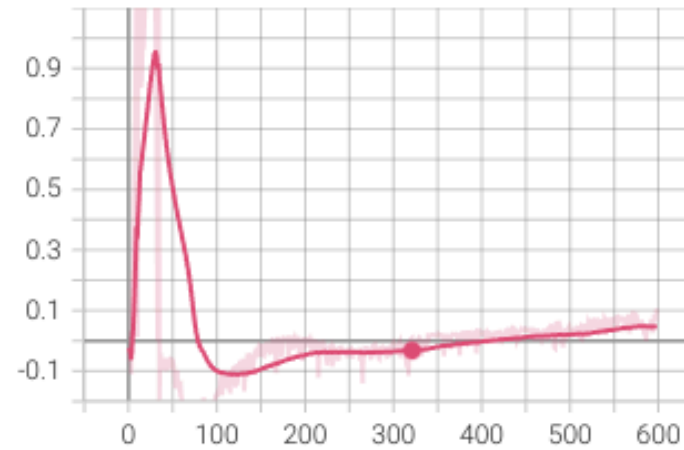
The learning process

Alice

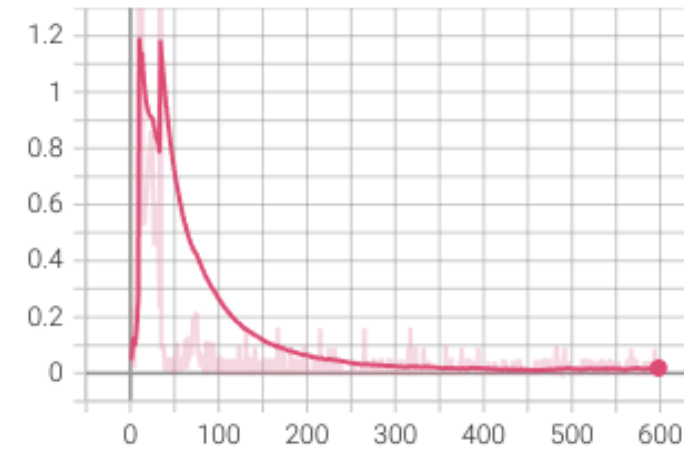
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Alice/rl_loss
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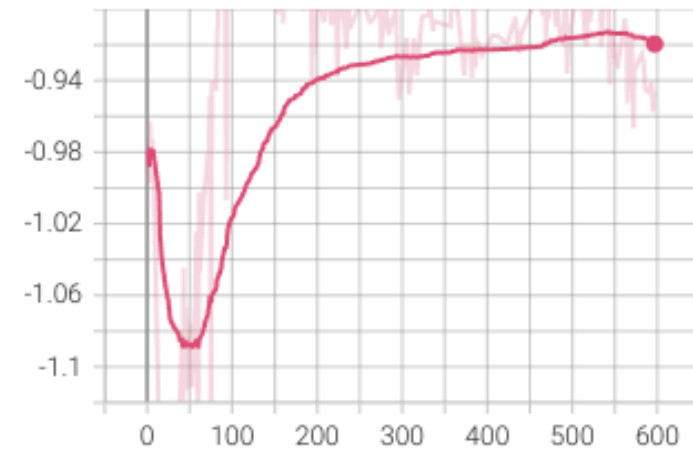


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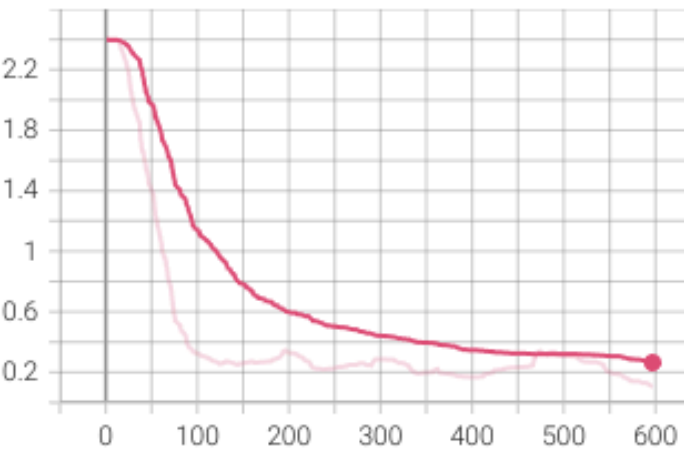


Bob

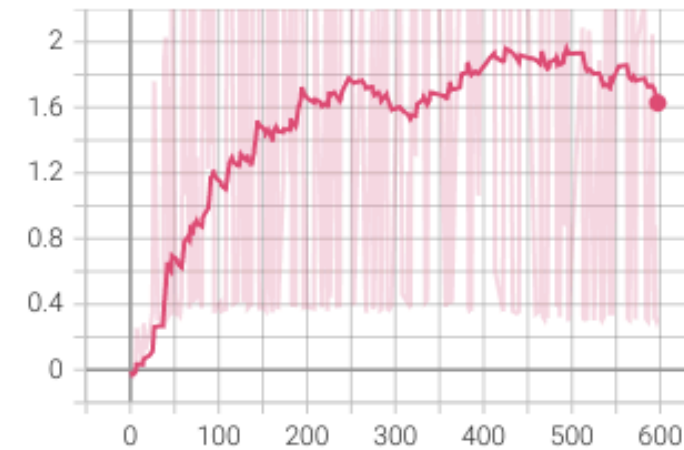
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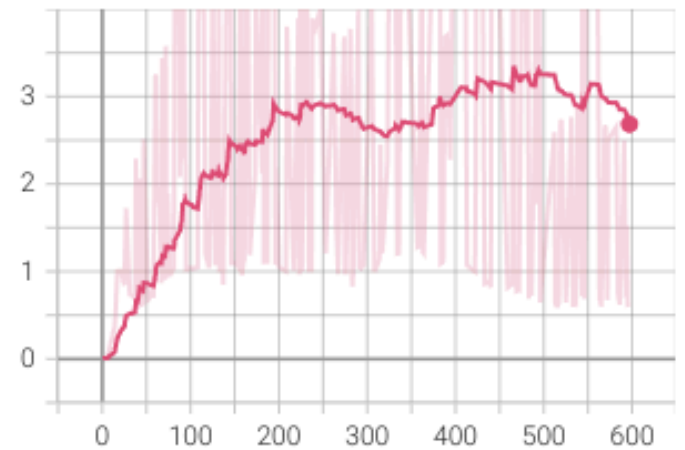
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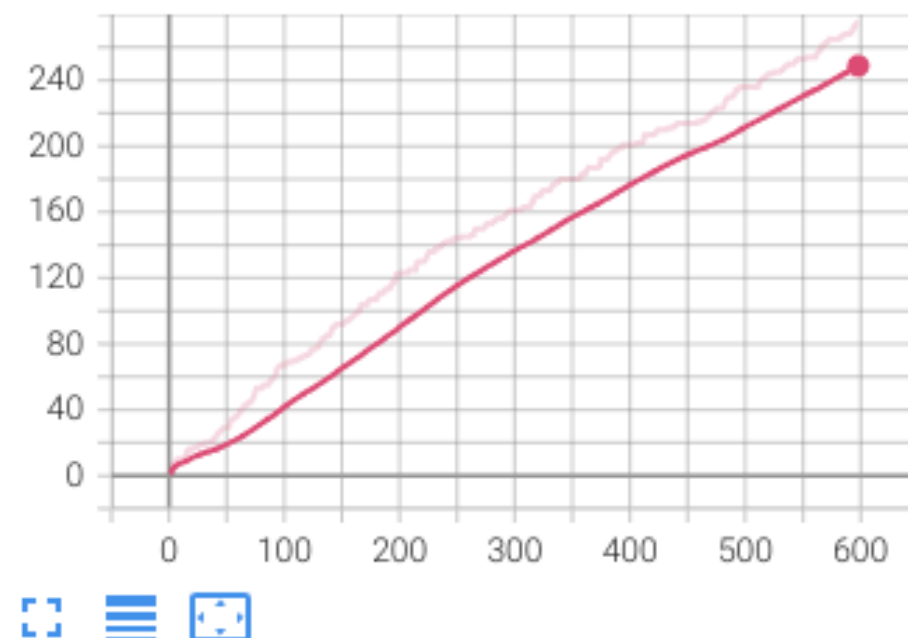


What have I achieved so far?

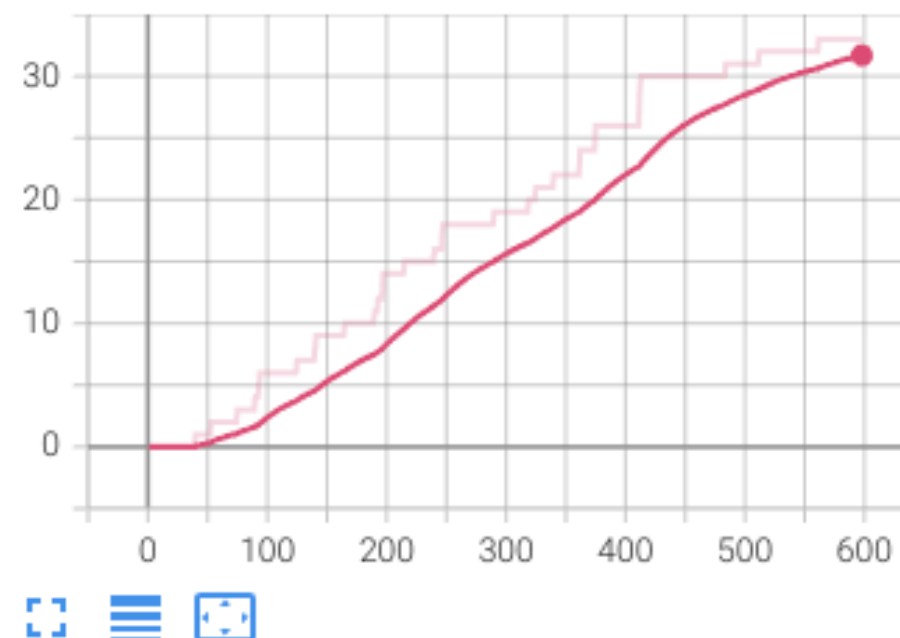
The **competition** between Alice and Bob is clear:

- Alice consistently proposes new challenges.
- Bob learns to solve them, getting better over time by “catching up” with Alice.

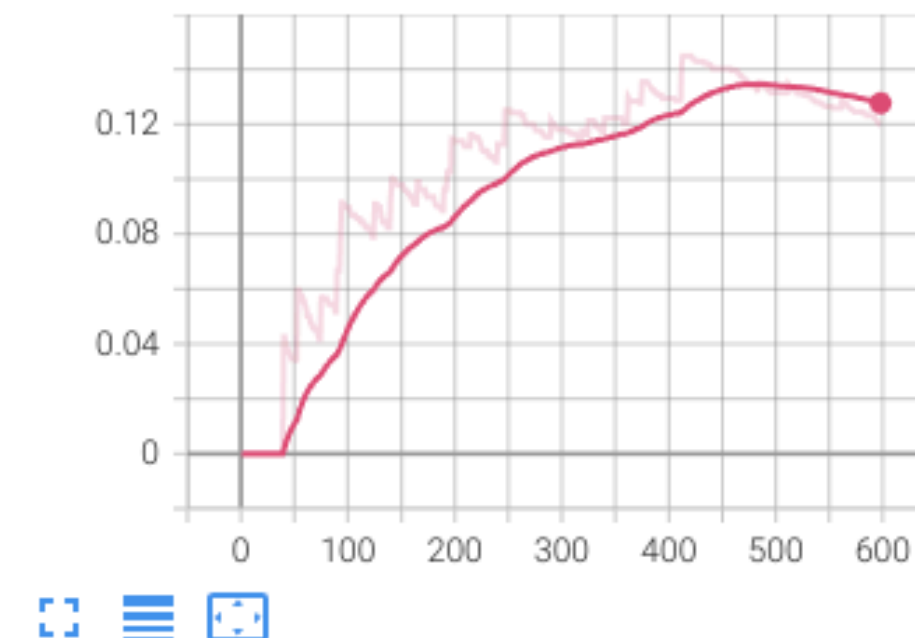
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Custom/CumulativeBobSuccesses
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Custom/Successes_vs_ValidGoals
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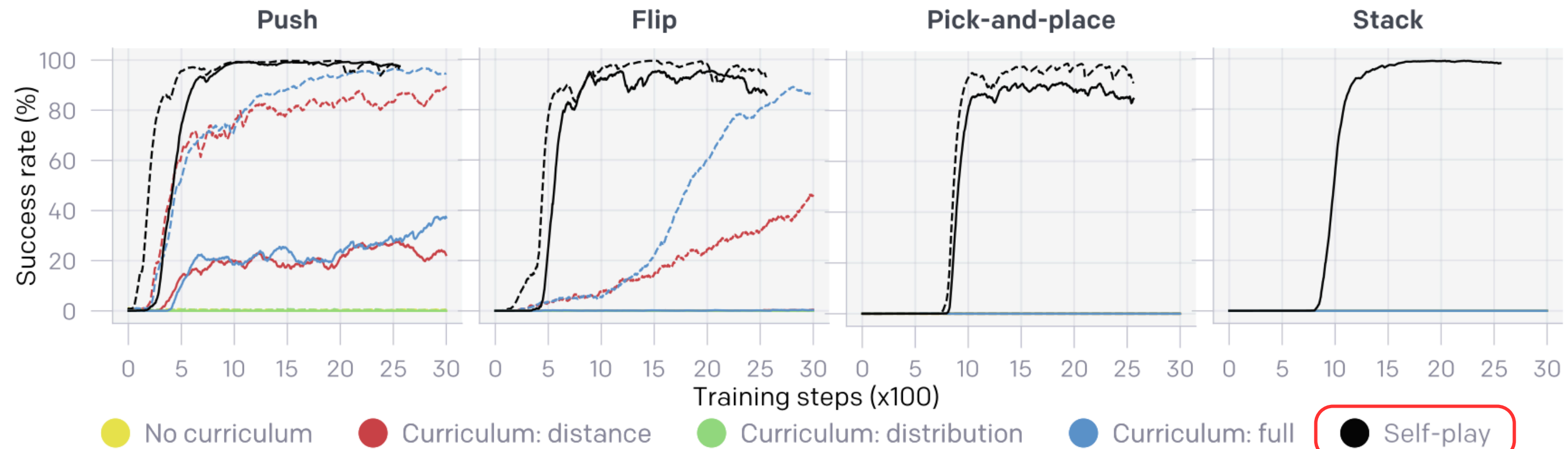


Eventually ...

... Bob learns to generalize.

Bob's policy is tested on a set of block manipulation tasks that were **never explicitly trained on**.

Push	Move blocks to desired poses.
Flip	Orient blocks to show a specific side.
Pick-and-place	Lift a block into the air.
Stack	Build a tower of two blocks.



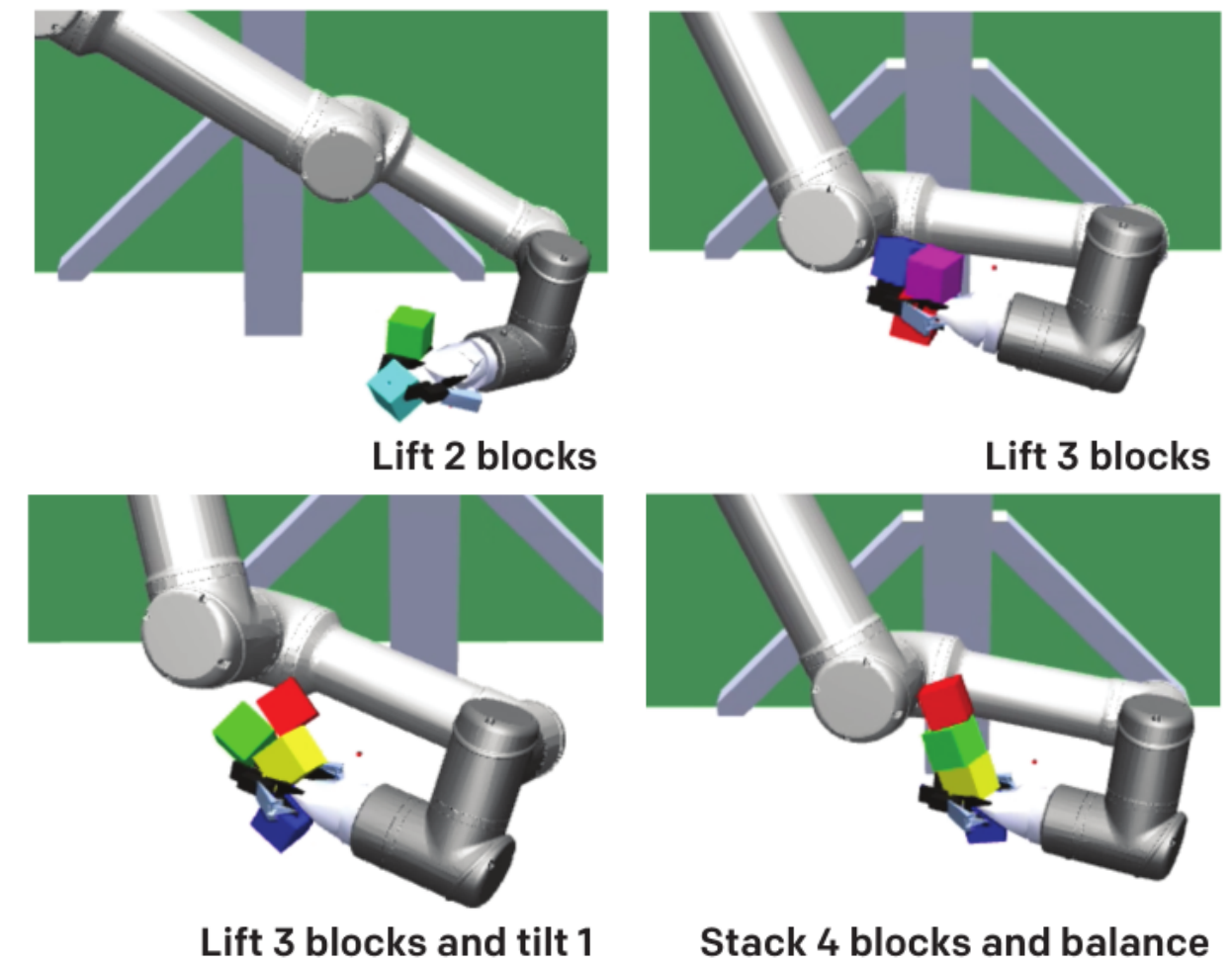
Discovering novel goals

The system is not limited by human-defined tasks.

Alice learns to challenge Bob in creative ways...

- Lifting 2 or 3 blocks simultaneously.
- Tilting a held block.
- Building and balancing a 4-block tower.

... and Bob learns creative ways to solve proposed goals!

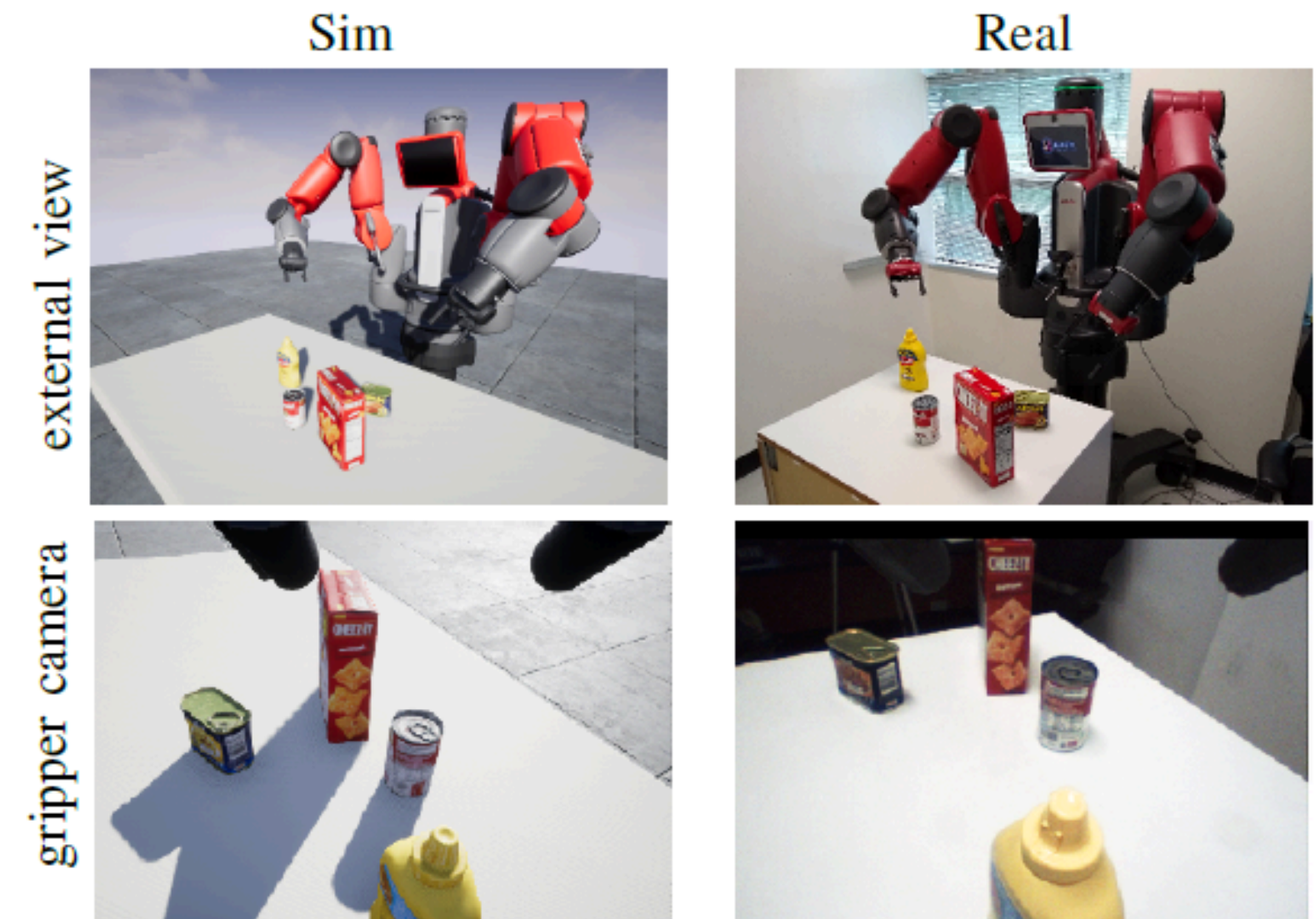


Limitations and future work

Sim2Real gap Setting the real world to an identical state for Bob is impossible.

Goal validity The goal discovery is not purely autonomous. We had to manually insert some rules to prevent Alice from cheating.

Future work Transfer Bob's learned policy to a physical robot using techniques like *domain randomization* and real-world fine-tuning.



https://research.nvidia.com/publication/2020-05_toward-sim-real-directional-semantic-grasping

Conclusion

Beyond surpassing manual curriculum baselines, Asymmetric Self-Play autonomously discovers innovative manipulation goals and solutions.

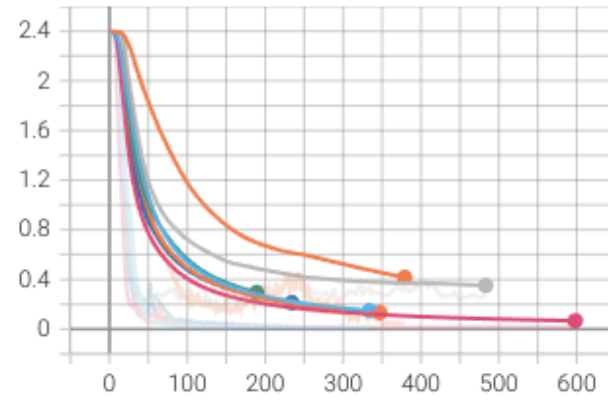
Summary

- Self-play generates a rich curriculum, outperforming hand-designed baselines.
- The learned policy solves challenging unseen tasks without explicit training on them (*zero-shot generalization*).
- The system discovers creative goals beyond what was manually specified.
- **ABC is critical** for making learning from sparse rewards possible.
- **Sim2Real gap** must be taken into account for real-world applications.

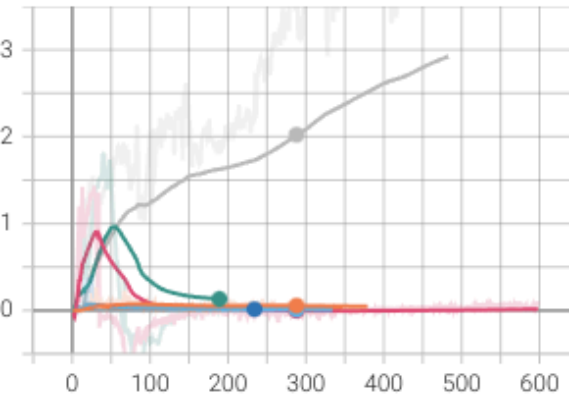
Bonus

Bloopers

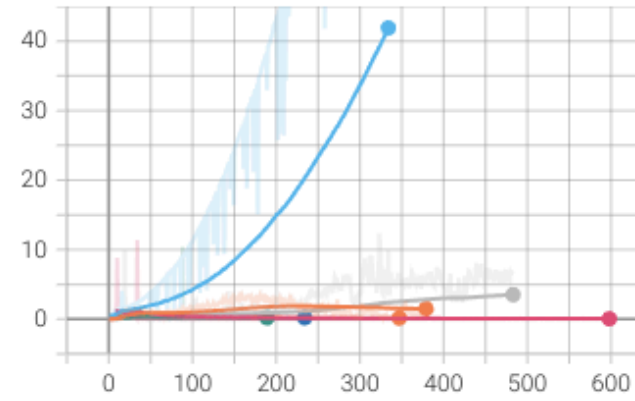
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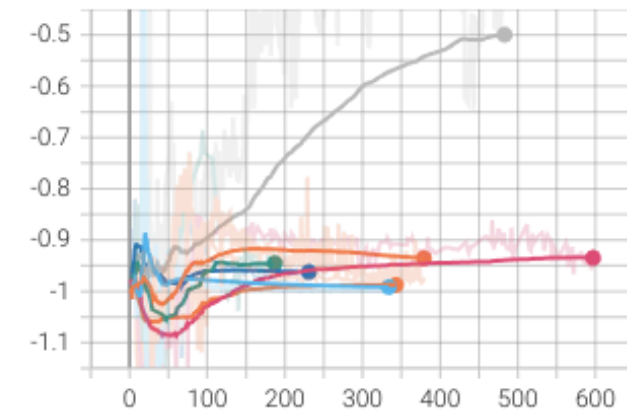
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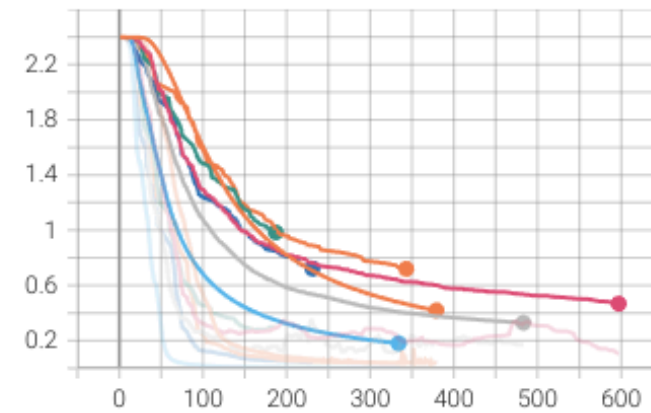
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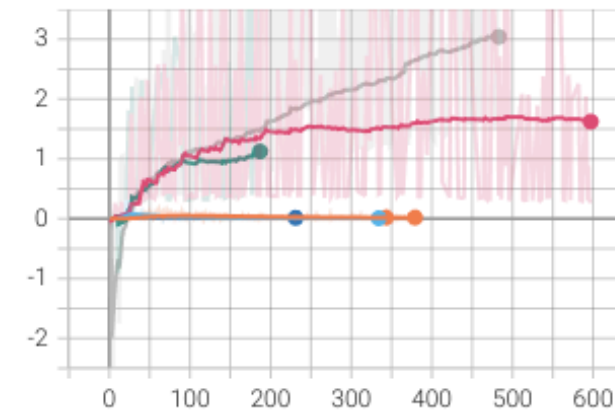
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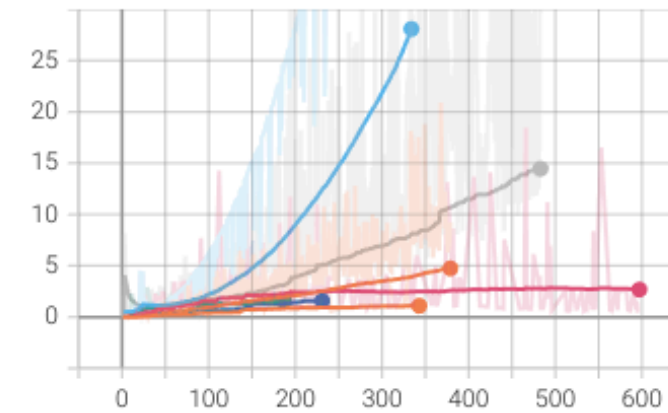
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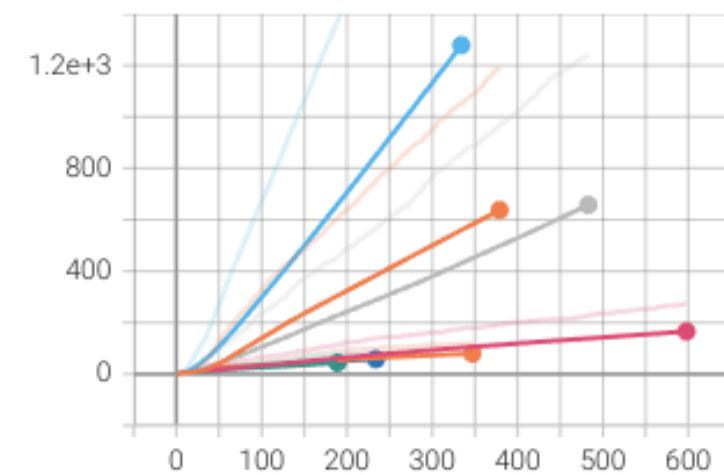
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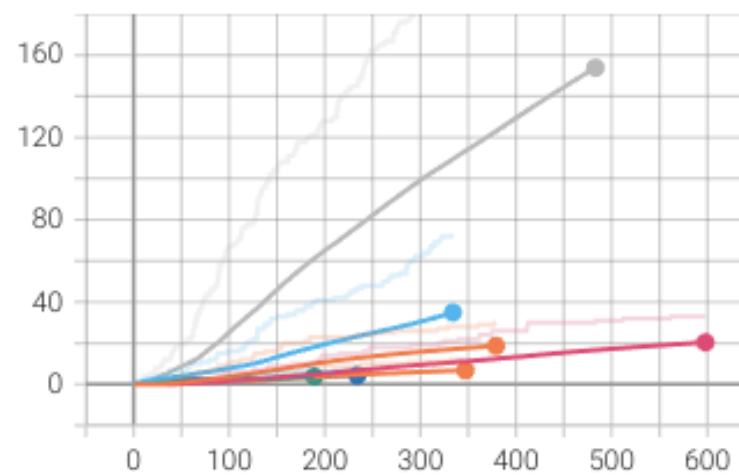
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